

Phase 1: Data Sourcing & Base Training

```
!pip install ultralytics opencv-python onnxruntime
```

```
Requirement already satisfied: ultralytics in /usr/local/lib/python3.12/dist-packages (8.4.56)
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (4.13.0.92)
Requirement already satisfied: onnxruntime in /usr/local/lib/python3.12/dist-packages (1.26.0)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.11.0+cu128)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.26.0+cu128)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.35.2)
Requirement already satisfied: ultralytics-thop>=2.0.18 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.18)
Requirement already satisfied: flatbuffers in /usr/local/lib/python3.12/dist-packages (from onnxruntime) (25.12.19)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from onnxruntime) (26.2)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from onnxruntime) (5.29.6)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultral)
Requirement already satisfied:ycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultral)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultra)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultra)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultral)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultral)
Requirement already satisfied: polars-runtime-32>=1.35.2 in /usr/local/lib/python3.12/dist-packages (from polars>=0.20.0->)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultral)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultra)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultra)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral) (3.29.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: setuptools>=42.0.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: cuda-toolkit>=12.8.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cusparselt-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral)
Requirement already satisfied: cuda-pathfinder~1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultral)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=1.8.0->ultral)
```

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
%cd /content/drive/MyDrive/Colab Notebooks/PPE
model.train(data="data.yaml", epochs=50, imgsz=640, batch=16, device=0)
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt> to 'yolov8n.pt': 100% ————— 6
/content/drive/MyDrive/Colab Notebooks/PPE
Ultralytics 8.4.56 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7
Overriding model.yaml nc=80 with nc=9

	from	n	params	module	arguments
0	-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1	-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2	-1	1	7360	ultralytics.nn.modules.block.C2f	[32, 32, 1, True]
3	-1	1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
4	-1	2	49664	ultralytics.nn.modules.block.C2f	[64, 64, 2, True]
5	-1	1	73984	ultralytics.nn.modules.conv.Conv	[64, 128, 3, 2]
6	-1	2	197632	ultralytics.nn.modules.block.C2f	[128, 128, 2, True]
7	-1	1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
8	-1	1	460288	ultralytics.nn.modules.block.C2f	[256, 256, 1, True]
9	-1	1	164608	ultralytics.nn.modules.block.SPPF	[256, 256, 5]

```
10         -1 1      0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
11     [-1, 6] 1      0 ultralytics.nn.modules.conv.Concat [1]
12         -1 1    148224 ultralytics.nn.modules.block.C2f [384, 128, 1]
13         -1 1      0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
14     [-1, 4] 1      0 ultralytics.nn.modules.conv.Concat [1]
15         -1 1    37248 ultralytics.nn.modules.block.C2f [192, 64, 1]
16         -1 1    36992 ultralytics.nn.modules.conv.Conv [64, 64, 3, 2]
17     [-1, 12] 1      0 ultralytics.nn.modules.conv.Concat [1]
18         -1 1   123648 ultralytics.nn.modules.block.C2f [192, 128, 1]
19         -1 1   147712 ultralytics.nn.modules.conv.Conv [128, 128, 3, 2]
20     [-1, 9] 1      0 ultralytics.nn.modules.conv.Concat [1]
21         -1 1   493056 ultralytics.nn.modules.block.C2f [384, 256, 1]
22    [15, 18, 21] 1   753067 ultralytics.nn.modules.head.Detect [9, 16, None, [64, 128, 256]]
Model summary: 130 layers, 3,012,603 parameters, 3,012,587 gradients, 8.2 GFLOPs
```

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed 
train: Fast image access  (ping: 6.5±13.2 ms, read: 0.3±0.3 MB/s, size: 94.1 KB)
train: Scanning /content/drive/MyDrive/Colab Notebooks/PPE/dataset/train/labels.cache... 230 images, 42 backgrounds, 0 corrupt
augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_avg')
val: Fast image access  (ping: 1.7±0.5 ms, read: 8.6±6.9 MB/s, size: 62.1 KB)
val: Scanning /content/drive/MyDrive/Colab Notebooks/PPE/dataset/valid/labels.cache... 62 images, 21 backgrounds, 0 corrupt
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum'
optimizer: AdamW(lr=0.000769, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to /content/drive/MyDrive/Colab Notebooks/PPE/runs/detect/train/labels.jpg...
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to /content/drive/MyDrive/Colab Notebooks/PPE/runs/detect/train
Starting training for 50 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
1/50	1.98G	1.69	4.349	1.588	16	640: 100%	17/17 1.6it/s 10.9s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 3/3 1.5it/s 2.
	all	83	230	0.0052	0.147	0.0115	0.00534
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
2/50	2.11G	1.586	3.251	1.416	16	640: 100%	17/17 3.4it/s 5.0s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 3/3 2.1it/s 1.

```
import os
model = YOLO("runs/detect/train/weights/best.pt")

metrics = model.val()

print(metrics.box.map)
print(metrics.box.map50)
```

Ultralytics 8.4.56  Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
Model summary (fused): 73 layers, 3,007,403 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access  (ping: 0.7±0.3 ms, read: 29.8±37.1 MB/s, size: 78.6 KB)
val: Scanning /content/drive/MyDrive/Colab Notebooks/PPE/dataset/valid/labels.cache... 62 images, 21 backgrounds, 0 corrupt:
Class Images Instances Box(P R mAP50 mAP50-95): 100% 6/6 1.2it/s 4.9s
all 83 230 0.258 0.239 0.232 0.123
Jacket 18 34 0.616 0.824 0.708 0.477
hard_hat 3 5 0 0 0 0
Gloves 15 31 0.211 0.161 0.201 0.102
Shoes 37 80 0.217 0.0125 0.046 0.0236
Hard_hat_required 11 21 0.537 0.714 0.654 0.308
Shoes_required 3 3 0 0 0.00218 0.000364
Jacket_required 21 40 0.202 0.075 0.138 0.0497
Gloves_required 8 16 0.282 0.125 0.108 0.0208
Speed: 8.9ms preprocess, 5.6ms inference, 0.0ms loss, 5.9ms postprocess per image
Results saved to /content/drive/MyDrive/Colab Notebooks/PPE/runs/detect/val-4
0.12273335024336951
0.23221538475934483

```
pt_size = os.path.getsize("runs/detect/train/weights/best.pt") / (1024 * 1024)
print("PT size:", round(pt_size, 2), "MB")
```

PT size: 5.96 MB

Phase 2: Edge Conversion & Quantization

```
model.export(format="onnx",half=True)
```

Ultralytics 8.4.56  Python-3.12.13 torch-2.11.0+cu128 CPU (Intel Xeon CPU @ 2.00GHz)

PyTorch: starting from 'runs/detect/train/weights/best.pt' with input shape (1, 3, 640, 640) BCHW and output shape(s) (1, 13)

ONNX: starting export with onnx 1.21.0 opset 20...

ONNX: slimming with onnxslim 0.1.94...

ONNX: converting to FP16...

ONNX: export success  1.9s, saved as 'runs/detect/train/weights/best.onnx' (5.9 MB)

Export complete (2.3s)

Results saved to /content/drive/MyDrive/Colab Notebooks/PPE/runs/detect/train/weights/best.onnx

Predict: yolo predict task=detect model=runs/detect/train/weights/best.onnx imgsz=640 half

Validate: yolo val task=detect model=runs/detect/train/weights/best.onnx imgsz=640 data=data.yaml half

Visualize: <https://netron.app>

'runs/detect/train/weights/best.onnx'

```
model = YOLO(
    "/content/drive/MyDrive/Colab Notebooks/PPE/runs/detect/train/weights/best.onnx"
)

metrics = model.val(
    data="/content/drive/MyDrive/Colab Notebooks/PPE/data.yaml"
)

print("mAP50-95:", metrics.box.map)
print("mAP50:", metrics.box.map50)

onnx_size = os.path.getsize(
    "/content/drive/MyDrive/Colab Notebooks/PPE/runs/detect/train/weights/best.onnx"
) / (1024 * 1024)

print("ONNX size:", round(onnx_size, 2), "MB")
```

Ultralytics 8.4.56  Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)

Loading /content/drive/MyDrive/Colab Notebooks/PPE/runs/detect/train/weights/best.onnx for ONNX Runtime inference...

WARNING  CUDA requested but CUDAEExecutionProvider not available. Using CPU...

Using ONNX Runtime 1.26.0 with CPUEExecutionProvider

Setting batch=1 input of shape (1, 3, 640, 640)

val: Fast image access  (ping: 0.4±0.1 ms, read: 27.8±24.3 MB/s, size: 68.8 KB)

val: Scanning /content/drive/MyDrive/Colab Notebooks/PPE/dataset/valid/labels.cache... 62 images, 21 backgrounds, 0 corrupt:

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95)
all	83	230	0.193	0.263	0.218	0.116
Jacket	18	34	0.546	0.824	0.709	0.496
hard_hat	3	5	0	0	0	0
Gloves	15	31	0.221	0.29	0.182	0.0879
Shoes	37	80	0	0	0.0341	0.0176
Hard_hat_required	11	21	0.483	0.762	0.585	0.249
Shoes_required	3	3	0	0	0.000794	0.000159
Jacket_required	21	40	0.186	0.1	0.144	0.0524
Gloves_required	8	16	0.11	0.125	0.0862	0.0274

Speed: 1.9ms preprocess, 102.2ms inference, 0.0ms loss, 2.6ms postprocess per image

Results saved to /content/drive/MyDrive/Colab Notebooks/PPE/runs/detect/val-7

mAP50-95: 0.11638571856323884

mAP50: 0.21755709971753182

ONNX size: 5.89 MB

Phase 3: The Inference Script & Metrics

```
import time
start = time.time()

for _ in range(100):
    model("/content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg")

fps = 100 / (time.time() - start)

print(fps)
```

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 41.2ms
Speed: 2.2ms preprocess, 41.2ms inference, 1.5ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 10.8ms
Speed: 2.7ms preprocess, 10.8ms inference, 2.0ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 6.1ms
Speed: 2.2ms preprocess, 6.1ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 6.0ms
Speed: 1.5ms preprocess, 6.0ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 6.0ms

```
Speed: 1.5ms preprocess, 6.0ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 5.9ms
Speed: 1.5ms preprocess, 5.9ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 5.9ms
Speed: 1.5ms preprocess, 5.9ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 5.8ms
Speed: 1.5ms preprocess, 5.8ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 5.8ms
Speed: 1.4ms preprocess, 5.8ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 6.1ms
Speed: 1.8ms preprocess, 6.1ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 5.9ms
Speed: 1.4ms preprocess, 5.9ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 6.7ms
Speed: 1.5ms preprocess, 6.7ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 5.8ms
Speed: 1.4ms preprocess, 5.8ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 5.9ms
Speed: 1.5ms preprocess, 5.9ms inference, 1.1ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 9.2ms
Speed: 1.9ms preprocess, 9.2ms inference, 1.9ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 5.9ms
Speed: 1.5ms preprocess, 5.9ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 448)

image 1/1 /content/drive/MyDrive/Colab Notebooks/PPE/dataset/test/images/p34.jpeg: 640x448 1 Gloves, 6.4ms
Speed: 1.5ms preprocess, 6.4ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 448)
```